

# An Application of Degree-Constrained Minimum Spanning Trees in Sensor Networks

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## Abstract

*In wireless sensor networks, it is an important task to periodically collect data from an area of interest for time-sensitive applications. The sensed data must be gathered and transmitted to a base station for further processing to meet the end-user queries. Since the network consists of low-cost nodes with limited battery power, it is a challenging task to design an efficient routing scheme that can minimize delay and offer good performance in energy efficiency, and long network lifetimes. In this paper, we propose a new routing protocol, called Degree-Constrained Minimal Spanning Tree (DCMST), to collect information efficiently. DCMST is efficient in the ways that it ensures maximal utilization of network energy, it makes the lifetime of the network longer, as well as it takes much lower time to complete a round. Simulation results show that DCMST achieves around 20% better performance than that of COSEN in respect of delay time. It also shows that DCMST can perform better than COSEN in total transmission distance, which implies the less total energy consumption. Therefore, our DCMST scheme gives a good compromise between energy efficiency and latency.*

## 1. Introduction

Wireless sensor networks are one of the most important technologies that will change the world [1] in that such networks can provide us with fine-granular observations about the physical world where we are living. Potential applications of wireless sensor networks include disaster rescue, energy management, medical monitoring, logistics and inventory management, and military reconnaissance, etc. With their capabilities for monitoring and control, the sensors are expected to be widely deployed. Such a network can provide a fine global picture through the

collaboration of many sensors with each observing a coarse local view [2], [3].

Among the various scopes one of the major applications of sensor network is to collect information periodically from a remote terrain where each node continually senses the environment and sends back this data to the base station (BS), which is usually located at considerably far from the target field [4], for further analysis. However, sensor networks are limited by the lifetime of the node's battery. Once they are deployed, the network can keep operating only until the battery power is sufficient. But it is almost impossible to replace the battery once deployed over an inaccessible terrain. Therefore, it is desirable that the network protocols should take care of issues like energy-efficiently, self-configuration, fault-tolerance, delay etc [5, 6]. Specially, energy efficiency is the most important issue in order to keep the network operating for longer time [7]. In order to save energy, it is useful to fuse the sensed raw data into more meaningful information before transmitting to the base station. This is because that, as sensors are deployed densely, it might generate huge redundant data and similar data from multiple nodes can be combined together so that the required number of transmission to the BS can be reduced. Another important issue in design consideration of a sensor network is data delivery time since in most cases data from sensor network are time critical as in the case of battle field or medical or security monitoring system. Such applications are required to receive the data from sensor nodes with minimum delay [6, 8].

In this paper, we propose DCMST, a hierarchical tree based protocol. Sensors are grouped into several clusters. In every cluster, a routing tree is constructed for data transmission. One sensor node is elected as a cluster head in every cluster based on the residual energy and this node remains as a cluster head for an optimal number of rounds. Among all cluster heads, a routing tree is also constructed. One cluster head is

selected to be the leader of all cluster heads based on some measures at every round. All nodes in a cluster send messages to the cluster head. Besides, all cluster heads send the information to leader of cluster heads. The leader is the node that transmits the information to the BS. After an optimal number of rounds, new group of cluster heads are selected. Due to the hierarchical tree structure, our protocol requires much lower time and energy as compared to other protocols of the wireless sensor networks for data collection.

The rest of the paper is organized as follows. In Section 2, we give an overview of the related routing protocols. The network and communication models of our proposed protocol are discussed in Section 3. A detail description of our approach is presented in Section 4. Section 5 shows a comparative analysis and some simulation results. Finally, Section 6 presents a concluding remark.

## 2. Related works

Among various proposed routing protocols the hierarchical protocols LEACH [9], PEGASIS [10], BCDP [11] and COSEN [12] provide elegant solutions to minimize energy consumption and to lengthen network lifetime. In LEACH, sensor nodes are organized into local clusters with one node in each cluster as cluster head. The cluster head receives data from all other sensors in the cluster, performs data aggregation, and transmits the aggregated data to the BS. LEACH uses rotation of the cluster head in order to evenly distribute the energy consumption. The operation of LEACH is organized into rounds. Each round begins with a set-up phase followed by a steady-state phase. During the set-up phase, each node will decide whether to become a cluster head or not according to a predefined criterion. After the selection procedure, the rest sensor nodes will decide to which cluster head it will belong for that round. Cluster head node then creates a TDMA schedule for all the number of nodes in the cluster. During the steady-state phase, each member node transmits to the associated cluster head within its assigned time slot. Cluster heads collect and aggregate all signals and then transmit the fused information to the remote BS. However, LEACH has some drawbacks. Firstly, the cluster setup overload that needs to be carried by the network at every round. Secondly, the complexity arises due to TDMA scheduling in transmission from the members to cluster head. Thirdly, there are many long distance transmissions in the network.

On the other hand, PEGASIS forms a chain including all nodes in the network. A chain is formed

by using a greedy algorithm so that each node can only communicate with its closest neighbor. In each round, a randomly selected node in the chain takes turn to transmit the aggregated information to the BS. PEGASIS saves energy by selecting only one leader node to transmit to the BS while other nodes transmit only to its local neighbor. However, it will cause excessive delay introduced by the distant node in a single chain.

BCDCP provides an improvement over LEACH where the energy expensive works such that cluster setup and routing path calculation are carried by the BS which has no energy limitation. COSEN provides a different solution without involving BS which means that the cluster setup and routing path calculation are all carried out by the sensor nodes themselves. COSEN operates in two phases – chain formation phase followed by data transmission phase. In the chain formation phase, several lower level chains are formed to include all sensor nodes. A node in a chain selects the nearest live node that is not already inserted into any other chain and adds it to the chain. Each chain is of fixed length. If the chain length exceeds a predefined length, new chain formation starts. This way of chain formation process continues until all the live nodes are grouped into chains. Then, in each lower level chain, one node is elected as a leader. All lower level leaders then connected as a higher level chain and one node is selected as the higher level leader. In data transmission phase, each sensor node sends the sensed information to its lower level leader. Then, each lower level leader sends the information towards higher level leader. Higher level leader sends the information to BS. Although COSEN outperforms in energy consumption and network delay than both LEACH and PEGASIS, it is still not good enough. In our work, we propose a hierarchical tree based protocol which will get better performance than COSEN both in energy consumption and network delay. Our proposal is completely self-organized and energy efficient with very limited delay.

## 3. Network and Communication models

In our proposed protocol, we consider the following network model assumptions:

- Data are transmitted periodically from the sensor network to the remote BS and delay critical.
- The BS is located far from the sensor network and fixed.
- All sensor nodes are homogeneous, energy constrained and immobile.

For the sake of uniformity, we use the same radio model as used in LEACH [9], PEGASIS [10] and COSEN [12]. We omit the details of energy consumed in transmitter amplifier for transmission and receiver electronics. They can be referred in [12].

#### 4. DCMST: Tree-based Sensor Network

The operation of our proposed Degree-Constrained Minimal Spanning Tree (DCMST) routing protocol can be divided by two phases: tree formation phase followed by data transmission phase. In the following sub-sections we discuss each of them in details.

##### 4.1. Tree Formation Phase

Sensor nodes are deployed randomly in the target field. Then, the network establishment begins with the formation of trees. Several lower level trees are formed to include all the sensor nodes. All nodes connected by a tree are treated as within a cluster and the tree among them is constructed by using a greedy algorithm. Within a cluster, one node is selected as cluster head by some criterions. After that a higher level tree is formed including all cluster head nodes. Among these cluster heads only one chosen head node sends information to BS.

At the beginning of tree formation only, we use the idea of minimum spanning tree (MST for short) in order to shorten the total transmission distance. This means that we construct a minimum spanning tree of the nodes in a cluster. It is possible that a node in the computed MST will be connected with many other nodes. In such case, this node needs to fuse more data collected from its neighbors than other nodes and consumes more energy. This may cause the node to die earlier than other nodes.

In order to avoid the situation that a node will be connected with many other nodes, we introduce the degree constraint to each tree node. For a connected, edge-weighted, and undirected graph, the degree of a node is the number of edges in which it participates. Given a positive integer  $d$ , the Degree-Constrained MST is a spanning tree with the smallest weight among all possible spanning trees which contain no nodes of degree greater than  $d$ . This problem is NP-hard, because the *Hamiltonian Path* problem (Problem ND1 in Garey and Johnson [13]), which is NP-complete, is a special case of  $d$ -MST with  $d = 2$  and all edge weights identical. For the cases that the vertices are points in the plane, and edge weights are the Euclidean distances between these points, Monma and Suri [14] showed that there always exists a MST with

degree no more than five. Paradimitriou and Vazirani [15] proved that finding a  $d$ -MST in the Euclidean plane is NP-hard when  $d = 3$ , and conjectured that it remains NP-hard when  $d = 4$ . It should be noticed that the chain in COSEN [12] is a tree (not necessary a MST) with no nodes of degree greater than 2.

In the rest of this section, we present the detailed implementation of the distributed protocol for constructing the degree constrained minimum spanning trees. The tree formation algorithm starts with the furthest node from BS. This furthest node is treated both as a tree node and a starting node. A node which is a starting node will broadcast a find-nearest-neighbor (FNN) message with largest transmission range to find the nearest live node. Once a node receives the FNN message, it sets a backoff timer of  $t_1$  seconds, where  $t_1$  is distributed in some range and depends on the signal strength of the received message. The more signal strength of the received message, the less  $t_1$  will be. When the timer expires, the node sends back an acknowledgement (ACK) message with its node identification to the starting node. If a node hears other ACK messages before its timer expires, it cancels its timer. When all tree nodes receive the ACK message, they will set a backoff timer of  $t_2$  seconds. Again,  $t_2$  is relative to the signal strength of the received ACK message. When  $t_2$  expires, the node sends a confirmation (CFM) message with node ID to inform the node sent the ACK message to be the next starting node and the link between them can be constructed. The above process will be repeated for finding next nearest live neighbor node until no live neighbors exist. If, during the process, the number of nodes in a cluster exceeds the predefined cluster size, the last starting node will broadcast a find-next-cluster (FNC) message to inform the nearest live neighbor node to be the starting node of the next cluster and the process proceeds for finding the nodes in next cluster.

As an example, Fig. 1 shows the process of constructing the degree-constrained MST for a network of five live nodes. At the beginning, node  $a$  is the starting node and it broadcasts a FNN message. Node  $b$  sends back an ACK message to node  $a$  since it is the nearest neighbor of node  $a$  (Fig. 1(b)). The link between nodes  $a$  and  $b$  is established and node  $b$  will be the next starting node. After node  $b$  sends a FNN message, node  $c$  will reply the ACK message and the link between nodes  $b$  and  $c$  will be established (Fig. 1(c)). After node  $c$  sends a FNN message, only nodes  $d$  and  $e$  can reply since nodes  $a$  and  $b$  are already in the tree. In this case, node  $d$  will reply an ACK message earlier than node  $e$  since it is closer to node  $c$ . At this time, nodes  $a$ ,  $b$  and  $c$  can hear the ACK message from node  $d$  and they will set a backoff timer for themselves

of a period of time according to the received signal strength from the ACK message sent by node  $d$ , respectively. In this case, the timer of node  $b$  will expire first and it sends a CFM message to node  $d$ . Therefore, node  $d$  will connect to node  $b$ , not to node  $c$  (Fig. 1(d)). The procedure will continue until all live neighbor nodes are found. Fig. 2(a) shows the final tree structure constructed by our procedure for the example in Fig. 1, compared with the chain structure constructed by COSEN as we shown in Fig. 2(b).

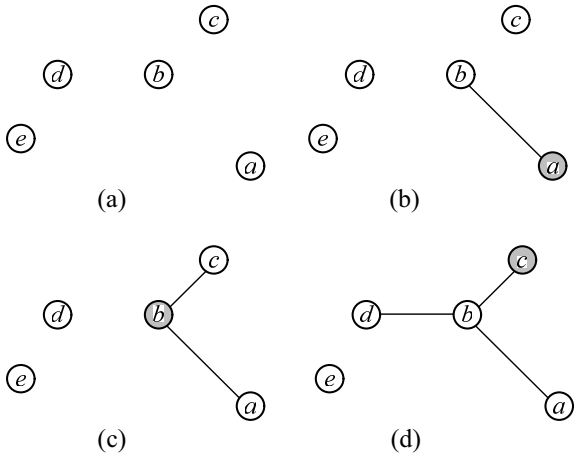


Fig. 1. The process of constructing the degree-constrained MST under our proposed scheme.

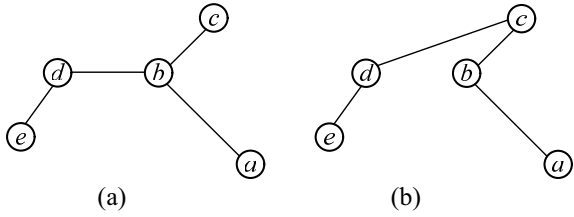


Fig. 2. The final tree and chain structures constructed by DCMST (a) and COSEN (b).

Our procedure for finding the degree constrained MST is simple. First, we let each node has a counter to indicate the current degree of itself. When a nearest live neighbor node replies with an ACK message to the starting node, some tree nodes will receive this message. Before a tree node can set a backoff timer, it has to check if its current degree counter is less than the degree limitation, say  $d$ . If its current degree counter equals to  $d$ , then it does not need to set a backoff timer to response the ACK message. When a link is established, the current degree counter will be incremented by one for both nodes connected by the link. When the above procedure stops, a degree

constrained spanning tree with no nodes of degree greater than  $d$  is obtained for a set of sensor nodes.

## 4.2. Data Transmission Phase

After the formation of tree and selection of cluster heads, sensors start data collection operation. At the beginning of the data collection and transmission phase each cluster head accumulates data from the member nodes within its cluster. For initiation of data transmission, we adapt the similar token passing mechanism as in PEGASIS [10]. If a node is elected as a cluster head it sends a token toward each end node of the tree. Each end node in a tree starts by transmitting data to the next node. The next node receives the data and fuses this data with its own and then transmits it to the next node. If an intermediate node has not received data from its all children nodes, it will wait till all data from its children nodes have been received. This is how data propagate from the end nodes to the cluster head in a tree. Every cluster head then transmits the information to the next cluster head in the higher level tree using the same fashion. Whenever the higher level cluster head gets all the information, it transmits the information to BS after data fusion.

## 5. Simulation Results

In this section, we evaluate the performance of our proposed routing protocol. We mainly compared DCMST with COSEN in our simulation. The simulation environment is described as follows. The simulation program is written by C++ programming language on .NET platform. In our simulation, we consider different network density from 100 nodes to 500 nodes. All nodes are placed randomly in a place of 50 meter \* 50 meter. Each sensor's location is represented by Cartesian coordinates. The base station is located at (0, 100). All nodes are divided evenly into 5 clusters. The simulation results are shown as the following figures, each representing an averaged summary over 100 runs. Fig. 3 shows the result of routing paths by using COSEN, a hierarchical chain-based protocol, on a sensor nodes distribution of a run from the experiment with 100 nodes in the network, while Fig. 4 shows the result of using our DCMST, a hierarchical tree-based protocol on the same sensor distribution in Fig. 3. It can be seen that our protocol performs better both in the delay time and total transmission distance than COSEN.

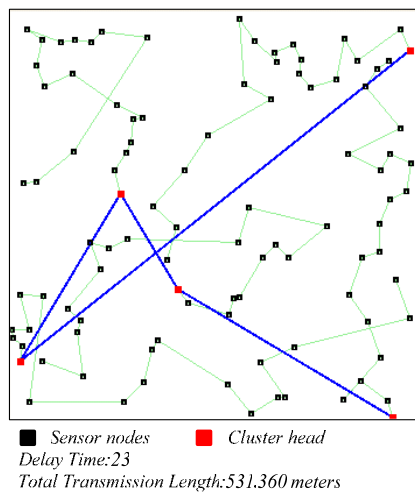


Figure 3. The routing path constructed by COSEN on a sensor distribution obtained from one run of the experiment with 100 nodes in the network.

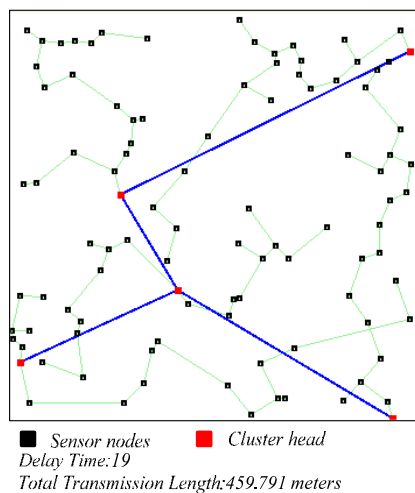


Figure 4. The routing path constructed by DCMST on the same sensor distribution in Figure 3.

Figures 5 and 6 show the simulation results of delay time and total transmission distance. As shown in Fig. 5, our DCMST performs better than COSEN in delay time about 21%. If we consider the total transmission distance, we found DCMST performs 15 percent better than COSEN. The improvement is achieved by using the tree-based structure instead of chain-based structure.

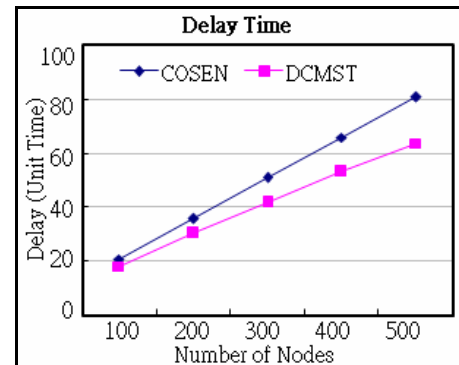


Figure 5. Delay time comparison for DCMST and COSEN.

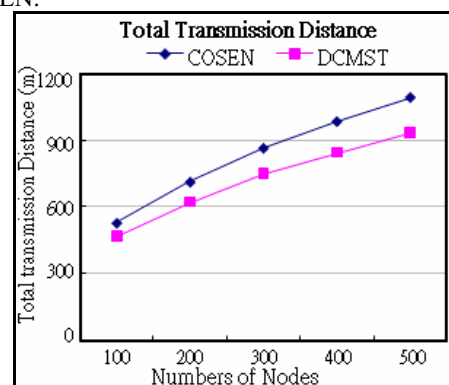


Figure 6. Total transmission distance comparison for DCMST and COSEN.

## 6. Conclusions

In this paper, we propose a new hierarchical tree-based routing protocol for efficiently collecting data in a sensor network. For designing the protocol, we consider how to reduce the delay of data collection and how to shorten the total transmission distance in order to reduce the energy consumption. Our protocol shows better performance than COSEN in terms of network delay and total transmission distance.

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